

Biology

Chapter 2: Cell

Welcome to AnithUncommon's Sample Resources!

In Biology , 2.1...

This sample will be divided into **three** parts: Introduction to Samples (this part), Lesson Objectives (What is Expected From Students), and most crucially, Syllabus Contents.

Note that the 'Syllabus Content' does NOT stand as a complete syllabus itself; rather, it is a tight compilation of notes gathered by Yassmin Younes to create a comprehensive sample for students.

If you want a complete version of these notes and think that you need help in Biology O'Level join our nonprofit!

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This sample is made thanks to, [Yassmin Younes](#)

— Made by students, for students.

Lesson Objectives:

At the end of the lesson, students should be able to...

- Recognize the specific roles of structures like the nucleus, mitochondria, ribosomes, and rough endoplasmic reticulum in maintaining cell life.
- Distinguish between the types of cells by their unique features such as the cellulose cell wall, chloroplasts, and large permanent vacuoles.
- Compare the simple design of a typical bacterium with the complex plant and animal cells.
- Apply the magnification formula ($I = A \times M$) to determine the actual size of microscopic specimens and convert accurately between millimetres (mm) and micrometres (μ).

Reflective Questions to Consider:

The cell is the basic building block of all living things, yet its design is never "one size fits all." Whether it is a microscopic bacterium surviving alone or a specialized human neurone working in a complex network, the internal architecture of a cell is perfectly tuned to its environment.

Below are the following core questions you will face as we explore this chapter:

- If the nucleus is the "brain" of the cell, how does a prokaryotic bacterium manage to grow, respire, and reproduce while lacking a nuclear envelope entirely?
- To what extent does the presence of a rigid cellulose cell wall in plants represent a trade-off between structural strength and the freedom of movement seen in animal cells?

- Specialized cells, such as red blood cells or xylem vessels, often lose their organelles or even their "living" status to function—at what point does a cell's specialization become more important than its own individual survival?

Syllabus Outline:

Unit 2: Cell

Tip!

- **The main thing is to understand how each structure, like the Mitochondria or Ribosomes, acts as a specific "department" in a factory that complements how the cell works as a whole. So, try to remember the cell not as a list of parts, but as a living machine: the Nucleus holds the blueprints, the Ribosomes build the products, and the Cell Membrane acts as the security guard.**

Activities in Class:

- **Lesson Breakdown**
- **Collaborate in drawing plant and animal cells with annotations**
- **Construct a comprehensive comparative table for plants and animals cell**
- **Hand-draw the bacterial cell and carefully label its organelles**

2.1 Animal and Plant Cells The Universal Parts (Found in Both) a. Cell Surface Membrane → A very thin layer of fat and protein. → Function: It is partially permeable; it controls what enters and leaves the cell, protecting the internal environment. b. Nucleus → The "Control Center" containing genetic information (DNA) in the form of chromosomes. → Function: It regulates all cell activities and is essential for cell division. c. Cytoplasm → A jelly-like substance that fills the cell. → Function: The site where most metabolic chemical reactions take place. It contains many small organelles. d. Mitochondria → Often called the "powerhouse." → Function: The site of Aerobic Respiration, where energy (ATP) is released from glucose. e. Ribosomes → Tiny structures often found on the Rough ER or floating freely.	Mentor's Notes: Try to RECOGNIZE that the Cell Wall and Cell Membrane are completely different structures. In exams, students often confuse them. Remember: Every cell has a membrane (the "security gate"), but <i>only</i> plants, fungi, and bacteria have a wall (the "outer armor"). When drawing a plant cell, always make the Vacuole the largest feature in the center.
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→ **Function:** The site of **Protein Synthesis** (where proteins are made).

The Plant-Only Parts (The Specialized Tools)

f. Cell Wall

→ A tough, non-living layer made of **cellulose** outside the cell membrane.

→ **Function:** Provides structural support and prevents the cell from bursting when it takes in water. It is fully permeable.

g. Chloroplasts

→ Green organelles containing the pigment **chlorophyll**.

→ **Function:** The site of **Photosynthesis**, where light energy is trapped to make food (glucose).

h. Large Permanent Vacuole

→ A large space filled with **cell sap**

→ **Function:** Helps keep the plant cell firm (turgid) by pushing against the cell wall.

In a real plant cell, this "lake" of sap pushes all the other organelles (like the nucleus) to the edges. This is a key diagnostic feature for your practical exams!

Cell sap is a solution of sugars and salts.

Note: This table highlights the primary diagnostic differences used in Paper 2 and Paper 4.

Comparison: Animal vs. Plant Cells

Feature	Animal Cell	Plant Cell
Shape	Irregular or flexible	Fixed and regular (due to cell wall)
Cell Wall	✗ Absent	✓ Present (Cellulose)
Chloroplasts	✗ Absent	✓ Present (in green parts)
Vacuole	Small and temporary (if present)	Large and permanent (central)
Energy Storage	Glycogen	Starch
Nucleus Position	Usually central	Usually pushed to the side

2.2 Bacterial Cells (Prokaryotes)

Key Features of a Bacterium

Bacterial cells are much smaller than plant or animal cells (usually 1–5 μm).

a. Cell Wall (Peptidoglycan)

→ A rigid outer layer that gives support. → **Function:** Unlike plants, this is made of a complex of proteins and sugars called **peptidoglycan**. It provides protection and structural support.

b. Cell Surface Membrane

→ Located just inside the cell wall. → **Function:** Controls the movement of substances into and out of the cell, just like in eukaryotes.

c. Cytoplasm

→ The "internal soup" of the cell. → **Function:** Contains the enzymes for metabolic reactions, though it lacks the complex compartmentalization found in animal cells.

d. Genetic Material (No Nucleus)

→ **Chromosomal DNA:** A single, circular loop of DNA found free-floating in the cytoplasm (the nucleoid).

→ **Plasmids:** Tiny, additional circles of DNA that carry extra genes (like resistance to medicine).

Try to **RECOGNIZE** the difference between **Chromosomal DNA** and **Plasmids**. In a bacterium, the **chromosomal DNA** is the "main hard drive" for survival, while **Plasmids** are like "extra USB drives" that carry **special instructions**, such as **antibiotic resistance**.

When labeling a diagram, be careful with the Cell Wall.

Remember that while plants use cellulose, bacteria use peptidoglycan.

Mentioning the specific material shows the examiner that you truly understand the chemical

e. Ribosomes (70S) → Smaller than those found in animal or plant cells (which are 80S).→ Function: The site of protein synthesis. f. Flagellum (s: flagella) → A whip-like "tail" (not present in all bacteria). → Function: Rotates like a propeller to allow the bacterium to move through liquids.	differences between Kingdoms! Bacteria's structure is "simpler" because they lack membrane-bound organelles. Is there a nucleus? No, Likely a bacterium. Is the DNA circular? Yes, Definitely a bacterium. Are there mitochondria? No, Definitely a bacterium. Are there plasmids? Yes, Definitely a bacterium.
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Comparison: Prokaryotic vs. Eukaryotic Cells

Feature	Bacterial Cell (Prokaryote)	Animal/Plant Cell (Eukaryote)
Nucleus	✗ Absent (DNA is free)	✓ Present (DNA inside)
DNA Shape	Circular	Linear
Mitochondria	✗ Absent	✓ Present
Ribosome Size	Smaller (70S)	Larger (80S)
Cell Wall	✓ Made of Peptidoglycan	✓ Cellulose (Plant) / ✗ (Animal)
Organelles	No membrane-bound parts	Many (Golgi, ER, Vacuoles)

2.3 Specialized Cells

I. Specialized Animal Cells

a. Ciliated Cells

→ **Location:** Lining the trachea and bronchi (respiratory tract).

→ **Feature:** Have tiny hair-like projections called **cilia**.

→ **Function:** The cilia beat in a rhythmic wave to move **mucus** (which traps dust and bacteria) upward and away from the lungs.

b. Neurones (Nerve Cells)

→ **Location:** Throughout the nervous system.

→ **Feature:** Extremely long with branched endings (dendrites) and a protective sheath.

→ **Function:** Designed to conduct **electrical impulses** quickly over long distances to coordinate body responses.

c. Red Blood Cells (Erythrocytes)

→ **Location:** In the blood of mammals.

→ **Feature:** Biconcave disc shape, **no nucleus**, and contains **haemoglobin**.

→ **Function:** To transport oxygen. The lack of a nucleus creates more space for haemoglobin.

d. Sperm and Egg Cells (Gametes)

→ **Location:** Testes (sperm) and Ovaries (eggs).

Try to **RECOGNIZE** the relationship between **Mitochondria** and **energy-hungry cells**.

If an exam question describes a cell with an unusually high number of mitochondria (like a Muscle cell or a Neurone), the answer is almost always that the cell requires a massive amount of energy (ATP) to perform its specialized function!

When studying Xylem vessels, remember that they are actually "dead" cells. They lose their cytoplasm and nucleus to become hollow tubes.

- **Sperm Feature:** Has a flagellum (tail) for swimming and many mitochondria for energy.
- **Egg Feature:** Much larger, contains a food store (yolk) for the developing embryo.
- **Function:** To fuse during fertilization to produce a zygote.

Hemoglobin's shape increases the surface area for oxygen diffusion.

II. Specialized Plant Cells

e. Root Hair Cells

- **Location:** Near the tips of growing roots.
- **Feature:** Has a long, thin outward extension (the "hair").
- **Function:** Increases the **surface area** of the root to maximize the absorption of water and mineral ions from the soil.

f. Xylem Vessels

- **Location:** In the stems, roots, and leaves.
- **Feature:** Long, hollow tubes with walls strengthened by **lignin**; they have no cytoplasm or nuclei.
- **Function:** To conduct water and mineral ions from the roots to the leaves and to provide structural support for the plant.

g. Palisade Mesophyll Cells

- **Location:** Just beneath the upper epidermis of a leaf.

→ **Feature:** Tall, column-shaped cells packed with a very high number of **chloroplasts**.

→ **Function:** The primary site of **photosynthesis**; their vertical shape allows many cells to be packed closely together to trap maximum sunlight.

Form vs. Function

Specialized Cell	Key Structural Feature	The "Why" (Function)
Red Blood Cell	No Nucleus	More room for Haemoglobin/Oxygen
Root Hair Cell	Long extension	Increased surface area for absorption
Ciliated Cell	Cilia (hairs)	Moving mucus out of the airways
Sperm Cell	Flagellum & Mitochondria	Swimming to the egg (Energy & Motion)
Xylem Vessel	Lignified walls	Strength and waterproof transport

2.4 Size of Specimens and Magnification

The Magnification Formula

To find the relationship between an image and its actual size, we use a simple triangular formula. You can rearrange this depending on what the question asks you to find.

The Formula Triangle: $I = A \times M$

- **I (Image Size):** The size of the drawing or the picture as you measure it with a ruler.
- **A (Actual Size):** The real size of the specimen in real life (often very tiny).
- **M (Magnification):** How many times larger the image is compared to the actual specimen.

Calculating the Variables

a. To find Magnification (M):

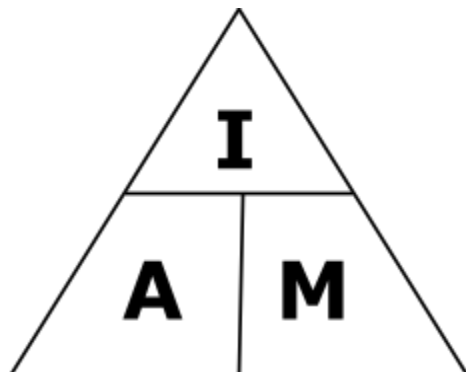
$$M = I / A$$

b. To find Actual Size (A):

$$A = I / M$$

c. To find Image Size (I):

$$I = A \times M$$



Try to **RECOGNIZE** the "Conversion Jump." Most students lose marks **not because of the math, but because they forget that 1 mm = 1000 micrometer**

Always convert your ruler measurements (usually in mm) into micrometres *before* you divide.

When using the magnification triangle, remember that Magnification (M) has no units. It is simply a "times" value (e.g., X400).

Magnify by " I AM " you get what is on top by multiplying the bottom values and get the bottom values by dividing the top value by the other bottom value.

Units and Conversions

In biology, we deal with things much smaller than a millimetre. The standard unit for cells is the **micrometer()**.

- **1 centimetre (cm)** = 10 millimetres (mm)
- **1 millimetre (mm)** = 1000 micrometres

The Conversion Workflow:

1. Measure the **Image (I)** using a ruler in **mm**.
2. Multiply that number by **1000** to get the size in
3. Proceed with your calculation.

Using a Scale Bar

Sometimes, a textbook doesn't tell you the magnification but gives you a **Scale Bar**

How to use it:

1. Measure the length of the scale bar with your ruler (this is your **I**).
2. Look at the number written above the scale bar (this is your **A**).
3. Divide the measured length (I) by the number on the bar (A) to find the magnification of the entire picture.

If your answer for magnification has "mm" next to it, you know you've made a mistake!

A scale bar is a small line at the bottom of the picture).

Step 1: Measure the image/line in mm.

Step 2: Convert mm to (multiply by 1000).

Step 3: Use the triangle to solve for the missing letter.

Step 4: Check if the answer makes sense (e.g., an actual cell should be a very small number!).

Mentors' Sources:

- <https://www.savemyexams.com/o-level/biology/cie/23/revision-notes/1-cells/1-1-cell-structure-and-function/animal-and-plant-cells/>
- <https://www.savemyexams.com/o-level/biology/cie/23/revision-notes/1-cells/1-1-cell-structure-and-function/animal-and-plant-cells-under-the-microscope/>
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- <https://www.savemyexams.com/o-level/biology/cie/23/revision-notes/1-cells/1-2-specialised-cells-tissues-and-organs/organisation-of-cells/>
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